

Objective

To implement and demonstrate ClassCastException in Java using try-catch by attempting to cast a String object into an Integer.

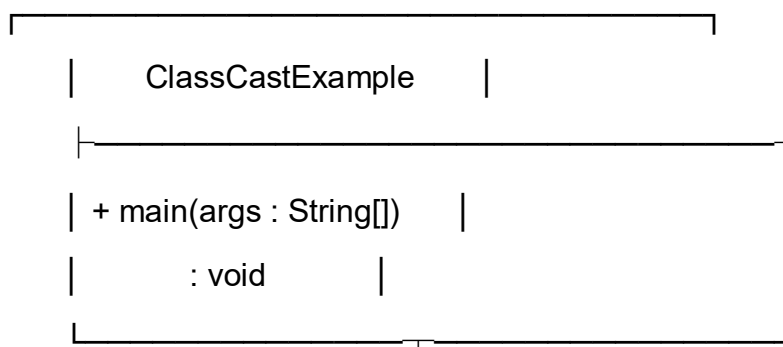
Steps for Execution

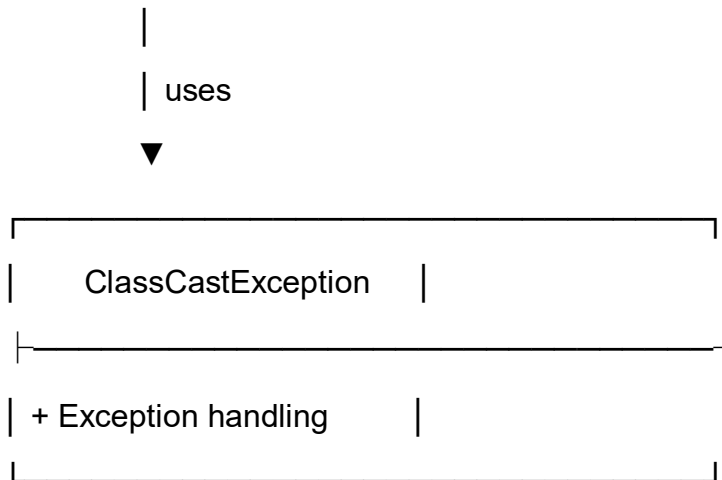
1. Create a Java project and package named javacore.
2. Create a file named ClassCastExample.java and enter the given code.
3. Store a String value in an Object reference.
4. Try to cast the String object into an Integer.
5. Handle the exception using catch.
6. Compile and run the program.
7. Observe the output.

Algorithm

1. Start the program.
2. Store "Hello Java" in an Object variable.
3. Try to cast the object into an Integer.
4. Catch ClassCastException.
5. Display the error message.
6. Display "Program ended.".
7. Stop the program.

UML Class Diagram





Relationship

- `ClassCastException` contains the `main()` method.
- The program handles `ClassCastException` using a try-catch block.
- The exception occurs because a `String` object cannot be directly cast to an `Integer`.

CODE:

```
package javacore;

public class ClassCastException {

    public static void main(String[] args) {

        Object value = "Hello Java";

        try {

            System.out.println("Value: " + value);

            Integer number = (Integer) value;

            System.out.println("Number: " + number);

        }

        catch (ClassCastException e) {
```

```
        System.out.println("Error: Cannot convert String into Integer.");
    }

    System.out.println("Program ended.");
}
}
```

Output

Value: Hello Java

Error: Cannot convert String into Integer.

Program ended.

Result

The program successfully demonstrates `ClassCastException` handling in Java. Since the value variable contains a String object, attempting to cast it to Integer causes a `ClassCastException`, which is handled using the catch block.